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Gilbert-Eyres; Matthew E. et al.

System and method for optimizing user pickup for an autonomous vehicle

Abstract

A method of operating an autonomous vehicle. The method includes collecting user data regarding a user of the autonomous vehicle and analyzing the user data to assign a pickup score to the user for determining when to relocate the autonomous vehicle from a first parking location. The autonomous vehicle is directed to relocate from the first parking location when the pickup score exceeds a predetermined threshold. A second parking location and a parking position are determined for the autonomous vehicle at a pickup location based on the user data when relocating the autonomous vehicle to pick up the user.

Inventors: Gilbert-Eyres; Matthew E. (Rochester Hills, MI), Wright; Adam L. (Livonia, MI), Neely; Matthew (Rochester, MI), Hosey; Eric T. (Rochester Hills, MI), Patenaude; Russell A. (Macomb Township, MI)

Applicant: GM GLOBAL TECHNOLOGY OPERATIONS LLC (Detroit, MI)

Family ID: 93381966

Assignee: GM Global Technology Operations LLC (Detroit, MI)

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Primary Examiner: Bendidi; Rachid

Assistant Examiner: Alqaderi; Nada Mahyoob

Attorney, Agent or Firm: Quinn IP Law

Background/Summary

INTRODUCTION

(1) The present disclosure relates to a system and a method for picking up a user of an autonomous vehicle.

(2) Autonomous vehicles include sensors, such as lidar, radar, and cameras, operable to detect vehicle operation and the environment surrounding the vehicle, and a computing device operable to control aspects of vehicle operation. Autonomous vehicles generally employ a vehicle navigation system integrated with vehicle controls, drive-by-wire systems, vehicle-to-vehicle communication, and/or vehicle-to-infrastructure technology to identify vehicle position and navigate the vehicle.

(3) Typically, a vehicle navigation system uses a global positioning system (GPS) to obtain its position data, which is then correlated to the vehicle's position relative to a surrounding geographical area. Based on the GPS signal, when directions to a specific waypoint are needed, routing to such a destination may be calculated, thereby determining a vehicle path. Specifically,

the vehicle sensors and the computing device may cooperate to identify intermediate way points and maneuver the vehicle between such way points to maintain the vehicle on the selected path. The vehicle may also need to locate parking while a user of the vehicle is running errands.

SUMMARY

(4) Disclosed herein is a method of operating an autonomous vehicle. The method includes collecting user data regarding a user of the autonomous vehicle and analyzing the user data to assign a pickup score to the user for determining when to relocate the autonomous vehicle from a first parking location. The autonomous vehicle is directed to relocate from the first parking location when the pickup score exceeds a predetermined threshold. A second parking location and a parking position are determined for the autonomous vehicle at a pickup location based on the user data when relocating the autonomous vehicle to pick up the user.

(5) Another aspect of the disclosure may be where directing the autonomous vehicle to relocate from the first parking location when the pickup score exceeds a predetermined threshold includes parking the autonomous vehicle at an intermediate parking location within a buffer parking zone prior to parking the autonomous vehicle at the pickup location.

(6) Another aspect of the disclosure may include directing the autonomous vehicle to relocate from the intermediate parking location to the second parking location when an updated parking score is assigned to the user that exceeds a predetermined pickup threshold.

(7) Another aspect of the disclosure may be where determining the second parking location and the parking position for the autonomous vehicle is based on identifying at least one user's need or at least one predetermined user preference.

(8) Another aspect of the disclosure may be where the at least one user's need is determined automatically by monitoring the user data collected.

(9) Another aspect of the disclosure may be where monitoring the user data collected includes monitoring recent purchases made by the user.

(10) Another aspect of the disclosure may be where analyzing the user data to assign the pickup score to the user includes monitoring for at least one time of purchase made by the user.

(11) Another aspect of the disclosure may be where analyzing the user data to assign the pickup score to the user includes monitoring calendar events of the user occurring within a predetermined range of time.

(12) Another aspect of the disclosure may be where analyzing the user data to assign the pickup score to the user includes monitoring travel plans of the user occurring within a predetermined range of time.

(13) Another aspect of the disclosure may be where analyzing the user data to assign the pickup score to the user includes monitoring a location of at least one personal electronic device associated with the user.

(14) Another aspect of the disclosure may be where the at least one personal electronic device includes a mobile device or a GPS enabled device.

(15) Another aspect of the disclosure may be where analyzing the user data to assign the pickup score to the user includes identifying a location of the user and determining an average time a visitor remains at the location.

(16) Another aspect of the disclosure may be where the location includes a place of commerce.

(17) Another aspect of the disclosure may be where determining the parking position for the autonomous vehicle includes tracking the user with at least one sensor on the autonomous vehicle to identify an object being transported by the user.

(18) Another aspect of the disclosure may be where the at least one sensor includes a camera.

(19) Disclosed herein is a non-transitory computer-readable storage medium embodying programmed instructions which, when executed by a processor, are operable for performing a method. The method includes collecting user data regarding a user of the autonomous vehicle and analyzing the user data to assign a pickup score to the user for determining when to relocate the

autonomous vehicle from a first parking location. The autonomous vehicle is directed to relocate from the first parking location when the pickup score exceeds a predetermined threshold. A second parking location and a parking position are determined for the autonomous vehicle at a pickup location based on the user data when relocating the autonomous vehicle to pick up the user.

(20) Disclosed herein is a system for an autonomous vehicle. The system includes a plurality of sensors and a controller in communication with the plurality of sensors. The controller is configured to direct the autonomous vehicle from a first parking location when a pickup score determined for a user of the autonomous vehicle exceeds a first predetermined threshold. The controller is also configured to direct the autonomous vehicle to a second parking location with a predetermined parking position at a pickup location based on user data collected.

(21) The above features and advantages, and other features and attendant advantages of this disclosure, will be readily apparent from the following detailed description of illustrative examples and modes for carrying out the present disclosure when taken in connection with the accompanying drawings and the appended claims. Moreover, this disclosure expressly includes combinations and sub-combinations of the elements and features presented above and below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of an example motor vehicle.

(2) FIG. 2 illustrates an example infrastructure with parking zones for the motor vehicle of FIG. 1.

(3) FIG. 3 illustrates an example flow for determining when to pick up a user of the motor vehicle of FIG. 1.

(4) FIG. 4 illustrates an example flow for determining a parking configuration for the motor vehicle of FIG. 1.

DETAILED DESCRIPTION

(5) Those having ordinary skill in the art will recognize that terms such as “above”, “below”, “upward”, “downward”, “top”, “bottom”, “left”, “right”, etc., are used descriptively for the figures, and do not represent limitations on the scope of the disclosure, as defined by the appended claims. Furthermore, the teachings may be described herein in terms of functional and/or logical block components and/or various processing steps. It should be realized that such block components may include a number of hardware, software, and/or firmware components configured to perform the specified functions.

(6) Referring to the FIGS., wherein like numerals indicate like parts referring to the drawings, wherein like reference numbers refer to like components, FIG. 1 shows a schematic view of a motor vehicle **10** positioned relative to a road surface, such as a vehicle lane **12**. As shown in FIG. 1, the vehicle **10** includes a vehicle body **14** enclosing a user **62**, a first axle having a first set of road wheels **16-1**, **16-2**, and a second axle having a second set of road wheels **16-3**, **16-4** (such as individual left-side and right-side wheels on each axle). Each of the road wheels **16-1**, **16-2**, **16-3**, **16-4** employs tires configured to provide fictional contact with the vehicle lane **12**. Although two axles, with the respective road wheels **16-1**, **16-2**, **16-3**, **16-4**, are specifically shown, nothing precludes the motor vehicle **10** from having additional axles.

(7) As shown in FIG. 1, a vehicle suspension system operatively connects the vehicle body **14** to the respective sets of road wheels **16-1**, **16-2**, **16-3**, **16-4** for maintaining contact between the wheels and the vehicle lane **12**, and for maintaining handling of the motor vehicle **10**. The motor vehicle **10** additionally includes a drivetrain **20** having a power-source or multiple power-sources **20A**, which may be an internal combustion engine (ICE), an electric motor, or a combination of such devices, configured to transmit a drive torque to the road wheels **16-1**, **16-2** and/or the road wheels **16-3**, **16-4**. The motor vehicle **10** also employs vehicle operating or control systems,

including devices such as one or more steering actuators **22** (for example, an electrical power steering unit) configured to steer the road wheels **16-1**, **16-2**, a steering angle (θ), an accelerator device **23** for controlling power output of the power-source(s) **20A**, a braking switch or device **24** for retarding rotation of the road wheels **16-1** and **16-2** (such as via individual friction brakes located at respective road wheels), etc.

(8) As shown in FIG. **1**, the motor vehicle **10** includes at least one sensor **25A** and an electronic controller **26** that cooperate to at least partially control, guide, and maneuver the vehicle **10** in an autonomous mode during certain situations. As such, the vehicle **10** may be referred to as an autonomous vehicle. To enable efficient and reliable autonomous vehicle control, the electronic controller **26** may be in operative communication with the steering actuator(s) **22** configured as an electrical power steering unit, accelerator device **23**, and braking device **24**. The sensors **25A** of the motor vehicle **10** are operable to sense the vehicle lane **12** and monitor a surrounding geographical area and traffic conditions proximate the motor vehicle **10**.

(9) The sensors **25A** of the vehicle **10** may include, but are not limited to, at least one of a Light Detection and Ranging (LIDAR) sensor, radar, and camera located around the vehicle **10** to detect the boundary indicators, such as edge conditions, of the vehicle lane **12**. The type of sensors **25A**, their location on the vehicle **10**, and their operation for detecting and/or sensing the boundary indicators of the vehicle lane **12** and monitor the surrounding geographical area and traffic conditions are understood by those skilled in the art, are not pertinent to the teachings of this disclosure, and are therefore not described in detail herein. The vehicle **10** may additionally include sensors **25B** attached to the vehicle body and/or drivetrain **20**.

(10) The electronic controller **26** is disposed in communication with the sensors **25A** of the vehicle **10** for receiving their respective sensed data related to the detection or sensing of the vehicle lane **12** and monitoring of the surrounding geographical area and traffic conditions. The electronic controller **26** may alternatively be referred to as a control module, a control unit, a controller, a vehicle **10** controller, a computer, etc. The electronic controller **26** may include a computer and/or processor **28**, and include software, hardware, memory, algorithms, connections (such as to sensors **25A** and **25B**), etc., for managing and controlling the operation of the vehicle **10**. As such, a method, described below and generally represented in FIG. **3**, may be embodied as a program or algorithm partially operable on the electronic controller **26**. It should be appreciated that the electronic controller **26** may include a device capable of analyzing data from the sensors **25A** and **25B**, comparing data, making the decisions required to control the operation of the vehicle **10**, and executing the required tasks to control the operation of the vehicle **10**.

(11) The electronic controller **26** may be embodied as one or multiple digital computers or host machines each having one or more processors **28**, read only memory (ROM), random access memory (RAM), electrically-programmable read only memory (EPROM), optical drives, magnetic drives, etc., a high-speed clock, analog-to-digital (A/D) circuitry, digital-to-analog (D/A) circuitry, and input/output (I/O) circuitry, I/O devices, and communication interfaces, as well as signal conditioning and buffer electronics. The computer-readable memory may include non-transitory/tangible medium which participates in providing data or computer-readable instructions. Memory may be non-volatile or volatile. Non-volatile media may include, for example, optical or magnetic disks and other persistent memory. Example volatile media may include dynamic random-access memory (DRAM), which may constitute a main memory. Other examples of embodiments for memory include a flexible disk, hard disk, magnetic tape or other magnetic medium, a CD-ROM, DVD, and/or other optical medium, as well as other possible memory devices such as flash memory.

(12) The electronic controller **26** includes a tangible, non-transitory memory **30** on which computer-executable instructions, including one or more algorithms, are recorded for regulating operation of the motor vehicle **10**. The subject algorithm(s) may specifically include an algorithm configured to monitor localization of the motor vehicle **10** and determine the vehicle's heading

relative to a mapped vehicle trajectory on a particular road course to be described in detail below.

(13) The motor vehicle **10** also includes a vehicle navigation system **34**, which may be part of integrated vehicle controls, or an add-on apparatus used to find travel direction in the vehicle. The vehicle navigation system **34** is also operatively connected to a global positioning system (GPS) **36** using an earth orbiting satellite. The vehicle navigation system **34** in connection with the GPS **36** and the above-mentioned sensors **25A** may be used for automation of the vehicle **10**. The electronic controller **26** is in communication with the GPS **36** via the vehicle navigation system **34**. The vehicle navigation system **34** uses a satellite navigation device (not shown) to receive its position data from the GPS **36**, which is then correlated to the vehicle's position relative to the surrounding geographical area. Based on such information, when directions to a specific waypoint are needed, routing to such a destination may be mapped and calculated. On-the-fly terrain and/or traffic information may be used to adjust the route. The current position of a vehicle **10** may be calculated via dead reckoning—by using a previously determined position and advancing that position based upon given or estimated speeds over elapsed time and course by way of discrete control points.

(14) The electronic controller **26** is generally configured, i.e., programmed, to determine or identify localization **38** (current position in the X-Y plane, shown in FIG. **1**), velocity, acceleration, yaw rate, as well as intended path **40**, and heading **42** of the motor vehicle **10** on the vehicle lane **12**. The localization **38**, intended path **40**, and heading **42** of the motor vehicle **10** may be determined via the navigation system **34** receiving data from the GPS **36**, while velocity, acceleration (including longitudinal and lateral g's), and yaw rate may be determined from vehicle sensors **25B**. Alternatively, the electronic controller **26** may use other systems or detection sources arranged remotely with respect to the vehicle **10**, for example a camera, to determine localization **38** of the vehicle relative to the vehicle lane **12**.

(15) As noted above, the motor vehicle **10** may be configured to operate in an autonomous mode guided by the electronic controller **26**. In such a mode, the electronic controller **26** may further obtain data from vehicle sensors **25B** to guide the vehicle along the desired path, such as via regulating the steering actuator **22**. The electronic controller **26** may be additionally programmed to detect and monitor the steering angle (θ) of the steering actuator(s) **22** along the desired path **40**, such as during a negotiated turn. Specifically, the electronic controller **26** may be programmed to determine the steering angle (θ) via receiving and processing data signals from a steering position sensor **44** (shown in FIG. **1**) in communication with the steering actuator(s) **22**, accelerator device **23**, braking device **24**.

(16) FIG. **2** illustrates an example parking infrastructure **60** having an entrance **70** and an exit **72** and being adjacent at least one buffer parking zone **64** and at least one pickup parking zone **66**. In the illustrated example, the buffer parking zone **64** functions as an intermediate parking area for the motor vehicle **10** between a main parking zone within the parking infrastructure **60** and the pickup parking zone **66** adjacent a point of interest **68**, such as a building or place of commerce. This configuration allows the motor vehicle **10** to transition from the parking infrastructure **60** to the buffer parking zone **64** for a length of time prior to the user **62** being ready for picked up by the motor vehicle **10** within the pickup parking zone **66**. By moving the motor vehicle **10** to the buffer parking zone **64**, delays caused by the motor vehicle **10** leaving the parking infrastructure **60** can be reduced.

(17) FIG. **3** illustrates an example method **100** for determining when to pick up the user **62** of the motor vehicle **10**. In the illustrated example, the method **100** begins at Block **102** and proceeds to Block **104** where the method **100** collects user data regarding the user **62** of the motor vehicle **10**. In one example, the user data can include calendared events, such as a start or an end time of an appointment, departure or arrival travel times, a geographic location of the user **62**, such as tracking the location of the user **62** relative to an exit of a building or a checkout area in a store.

(18) The user **62** can be tracked by monitoring a GPS position of a mobile device or GPS enabled device, such as a smart watch, associated with the user **62**. The user data is continuously collected

to provide the method **100** with the most current activities of the user **62** to provide insight into when the user **62** may be ready for pickup. The method **100** can also collect the user data through infrastructure (V2X) communications, ultra-wide band (UWB) communications, or obtain the user data from another source, such as over the internet.

(19) Further, the user data collected can include historic travel patterns of the user **62** or an average arrival time or departure time for the user **62** from a given location. Alternatively, the user data could include recent transactions made by the user **62**, such as making an electronic payment to purchase an item, which would indicate an approaching pickup time for the user **62**. Additionally, the method **100** can obtain crowd sourced information regarding a location of the user **62**, such as an average time a visitor remains at the location.

(20) Once some user data has been collected at Block **104** relevant to the user **62** needing to be picked up by the autonomous vehicle **10**, the method **100** then proceeds to Block **106** for analyzing the user data and assigning a pickup score for the user **62** based on the user data currently collected. The user data is analyzed to determine when to direct the motor vehicle **10** to relocate from the parking infrastructure **60**.

(21) As the method **100** continuously collects the user data from Block **104**, Block **106** analyzes the data to assign a current or updated pickup score for the user. In one example, if the user data analyzed indicates the user **62** recently made a purchase, a higher score pickup score is assigned for the user **62** that may indicate that a time to pick up the user **62** is approaching. With the pickup score calculated, the method **100** proceeds to Block **108** to determine if the motor vehicle **10** should be directed to relocate based on the pickup score.

(22) At Block **108**, the method **100** compares the pickup score for the user **62** to a first predetermined range of pickup scores. In the illustrated example, the first predetermined range of pickup scores includes scores greater than a pickup score of X and less than a pickup score of Y. If the pickup score assigned from Block **106** is within the first predetermined range of pickup scores, then the method **100** will proceed to Block **110**.

(23) At Block **110**, the motor vehicle **10** is directed to relocate from the parking infrastructure **60** or a first parking location, to an intermediate location in the buffer parking zone **64** in anticipation of the user **62** being ready for pickup at a pickup location within the pickup zone **66**. At this point, the method **100** returns to Block **106** for further analysis of user data to generate an updated or current pickup score for determining how to direct the motor vehicle **10** for picking up the user **62**.

(24) With the current or updated pickup score, the method **100** proceeds to Block **108** to compare the pickup score to the first predetermined range of pickup scores. If the updated pickup score exceeds the first predetermined range of pickup scores in Block **108**, the method **100** proceeds to Block **112**.

(25) At Block **112**, the method **100** compares the current or updated pickup score to a second predetermined range of pickup scores. In the illustrated example, the second predetermined range of pickup scores includes scores greater than a pickup score of Y. The second predetermined range of pickup scores indicates that a status of the user **62** is ready for pickup such that the motor vehicle **10** should be directed to relocate to a parking location within the pickup parking zone **66**. The method **100** then proceeds to Block **114** to direct the motor vehicle **10** to relocate from either the parking infrastructure **60** or the buffer parking zone **64** to the pickup location within the pickup zone **66**. While the above examples direct the motor vehicle **10** to relocate to the buffer parking zone **64** based on the initial pickup score, the motor vehicle **10** can proceed from the parking infrastructure **60** directly to the pickup location within the pickup parking zone **66** without relocating to an intermediate parking location within the buffer parking zone **64**. Before the motor vehicle **10** can be relocated to the pickup parking zone **66**, the parking location and positioning of the motor vehicle **10** within the pickup parking zone **66** should be determined.

(26) With the pickup status granted for the user **62**, the parking location and position of the motor vehicle **10** is determined as illustrated by the method **200** shown in FIG. **4**. Determining the

parking location and position for the motor vehicle **10** with the method **200** begins at Block **202** and proceeds to Block **204** with a confirmation of the pickup status being granted as discussed above.

(27) The method **200** then determines if the user **62** created preferences or pre-determined options for parking to implement within the pickup parking zone **66** at Block **206**. For example, the preferences or predetermined options could include parking in a space within a predetermined distance from a store exit or providing additional room in the parking space for one side of the motor vehicle **10** over the other. Once the preferences or predetermined options are determined, the method **200** proceeds to Block **208** to determine if the user has additional needs.

(28) At Block **208**, the user data is further analyzed to identify parking needs of the user **62**. In particular, the method **200** can analyze purchase records of the user **62** to determine a size or type of item purchased. The method **200** can also determine if the user **62** may have ADA (Americans with Disabilities Act) needs that require a specific parking space or orientation of the motor vehicle **10** at the pickup location at Block **210**.

(29) The method **200** then proceeds to Block **212** to attempt to automatically identify parking needs of the user **62** based on the user data. If the method **200** can automatically identify the parking needs at Block **212**, the method **200** proceeds to Block **214** to assign a predetermined pickup style based on the automatically identified needs. In one example, the predetermined pickup style includes a parking location and position for the motor vehicle **10**.

(30) Alternatively, if the method **200** cannot automatically identify the parking needs, the user **62** can select a predetermined parking position from a number of predetermined options, such as Block **216** (grocery), Block **218** (large item), Block **220** (lumber), or a customized parking position in Block **222**. In the case of lumber or large items, the predetermined parking position for the motor vehicle **10** may include a parking location near an exit of the building and a position of the motor vehicle **10** that allows for trunk access towards the building to avoid the user **62** having to walk around the motor vehicle **10**.

(31) The cameras on the motor vehicle **10** can also be used to identify objects being transported by the user **62** while the user **62** is approaching the motor vehicle **10** to determine a custom parking position for the user **62**. For example, if the cameras associated with the sensors **25A**, **25B** determines that the user **62** is carrying a large package, a parking position will be selected that accommodates placing the package in the motor vehicle **10**, such as reversing into the pickup location or opening the trunk as the user **62** approaches. The cameras can also work in connection with the mobile device or GPS enabled device associated with the user **62** to improve tracking of the user **62** approaching the motor vehicle **10**.

(32) The method **100** can then proceed to Block **116** once the parking location and position are determined and the motor vehicle **10** is parked at Block **224** to end the methods.

(33) The detailed description and the drawings or figures are supportive and descriptive of the disclosure, but the scope of the disclosure is defined solely by the claims. While some of the best modes and other embodiments for carrying out the claimed disclosure have been described in detail, various alternative designs and embodiments exist for practicing the disclosure defined in the appended claims. Furthermore, the embodiments shown in the drawings, or the characteristics of various embodiments mentioned in the present description are not necessarily to be understood as embodiments independent of each other. Rather, it is possible that each of the characteristics described in one of the examples of an embodiment may be combined with one or a plurality of other desired characteristics from other embodiments, resulting in other embodiments not described in words or by reference to the drawings. Accordingly, such other embodiments fall within the framework of the scope of the appended claims.

Claims

1. A method of operating an autonomous vehicle, the method comprising: assigning an initial pickup score to a user of the autonomous vehicle for determining when to relocate the autonomous vehicle from a first parking location is based on analyzing user data collected regarding a location of the user relative to at least one of an exit of a current location of the user or a checkout area of the current location; controlling the autonomous vehicle to relocate from the first parking location when the initial pickup score exceeds an initial predetermined threshold; and determining a second parking location and a parking position for the autonomous vehicle at the second parking location based on the user data when relocating the autonomous vehicle to pick up the user.
2. The method of claim 1, wherein controlling the autonomous vehicle to relocate from the first parking location when the initial pickup score exceeds the initial predetermined threshold includes controlling the autonomous vehicle to relocate to an intermediate parking location within a buffer parking zone prior to controlling the autonomous vehicle to relocate from the intermediate parking location to the second parking location.
3. The method of claim 2, wherein controlling the autonomous vehicle to relocate from the intermediate parking location to the second parking location is based on an updated pickup score assigned to the user that exceeds an updated predetermined threshold with the updated pickup score based on a recent purchase by the user when the autonomous vehicle is located at either the first parking location or after the autonomous vehicle has relocated from the first parking location.
4. The method of claim 1, wherein determining the second parking location and the parking position at the second parking location for the autonomous vehicle is based on identifying at least one predetermined user preference.
5. The method of claim 1, wherein determining the second parking location and the parking position at the second parking location for the autonomous vehicle is based on identifying a loading requirement for the autonomous vehicle with the loading requirement being determined automatically by monitoring recent purchases by the user when the autonomous vehicle is located at either the first parking location or after the autonomous vehicle has relocated from the first parking location.
6. The method of claim 1, wherein analyzing the user data to assign the initial pickup score to the user includes monitoring for at least one time of purchase made by the user when a personal electronic device associated with the user is located in a place of commerce determined by a GPS location of the personal electronic device and the at least one time of purchase occurs when the autonomous vehicle is located at either the first parking location or after the autonomous vehicle has relocated from the first parking location.
7. The method of claim 1, wherein analyzing the user data to assign the initial pickup score to the user includes monitoring calendar events of the user occurring within a predetermined range of time.
8. The method of claim 1, wherein analyzing the user data to assign the initial pickup score to the user includes monitoring travel plans of the user occurring within a predetermined range of time.
9. The method of claim 1, wherein analyzing the user data to assign the initial pickup score to the user includes monitoring the current location of at least one personal electronic device associated with the user.
10. The method of claim 9, wherein the at least one personal electronic device includes a mobile device or a GPS enabled device.
11. The method of claim 1, wherein analyzing the user data to assign the initial pickup score to the user includes determining an average time a visitor remains at the current location.
12. The method of claim 11, wherein the location includes a place of commerce.
13. The method of claim 1, wherein determining the parking position for the autonomous vehicle includes tracking the user with at least one sensor on the autonomous vehicle to identify an object being transported by the user.

14. The method of claim 13, wherein the at least one sensor includes a camera.

15. The method of claim 1, wherein assigning the initial pickup score to the user is based on analyzing user data collected regarding a recent purchase by the user when the autonomous vehicle is located at either the first parking location or after the autonomous vehicle has relocated from the first parking location.

16. A non-transitory computer-readable storage medium embodying programmed instructions which, when executed by a processor, are operable for performing a method comprising: assigning an initial pickup score to a user of an autonomous vehicle for determining when to relocate the autonomous vehicle from a first parking location is based on analyzing user data collected regarding a relative location of the user relative to at least one of an exit of a current location of the user or a checkout area of the current location; controlling the autonomous vehicle to relocate from the first parking location when the initial pickup score exceeds an initial predetermined threshold; and determining a second parking location and a parking position for the autonomous vehicle at the second parking location based on the user data when relocating the autonomous vehicle to pick up the user.

17. The storage medium of claim 16, wherein controlling the autonomous vehicle to relocate from the first parking location when the initial pickup score exceeds the initial predetermined threshold includes controlling the autonomous vehicle to relocate to an intermediate parking location within a buffer parking zone prior to controlling the autonomous vehicle to relocate from the intermediate parking location to the second parking location.

18. The storage medium of claim 17, wherein controlling the autonomous vehicle to relocate from the intermediate parking location to the second parking location is based on an updated pickup score is assigned to the user that exceeds an updated predetermined threshold with the updated pickup score is based at least partially on a recent purchase by the user when the autonomous vehicle is located at either the first parking location or after the autonomous vehicle relocated from the first parking location.

19. A system for an autonomous vehicle comprising: a plurality of sensors; a controller in communication with the plurality of sensors, the controller configured to: assign an initial pickup score to a user of an autonomous vehicle for determining when to relocate the autonomous vehicle from a first parking location is based on analyzing user data collected regarding a relative location of the user relative to at least one of an exit of a current location of the user or a checkout area of the current location; control the autonomous vehicle to relocate from a first parking location when an initial pickup score exceeds an initial predetermined threshold; and determine a second parking location and a parking position for the autonomous vehicle at the second parking location based on the user data when relocating the autonomous vehicle to pick up the user.

20. The system of claim 19, wherein the controller is configured to control the autonomous vehicle to relocate from the first parking location when the initial pickup score exceeds the initial predetermined threshold by controlling the autonomous vehicle to relocate to an intermediate parking location within a buffer parking zone prior to controlling the autonomous vehicle to relocate to the second parking location based on an updated pickup score assigned to the user that exceeds an update predetermined threshold with the updated pickup score based at least partially on a recent purchase by the user when the autonomous vehicle is located at either the first parking location or after the autonomous vehicle relocated from the first parking location.
